Appendix: Complete R Code for FINSTAD Case 1

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SISON, Aaron Joshua Estacio

```r
# Load necessary libraries
library(dplyr)
library(tidyr)
library(lubridate)
library(DBI)
library(RSQLite)
library(ggplot2)
library(scales)

# Set working directory to project root so data paths resolve consistently
# Ensure data paths resolve correctly regardless of working directory
if (dir.exists("data")) {
  data_dir <- "data"
} else if (dir.exists("../data")) {
  data_dir <- "../data"
} else {
  stop("Cannot find data/ directory. Run from project root or scripts/.")
}

# Global options
options(stringsAsFactors = FALSE, scipen = 999, width = 72)

# Helper to parse BDO volume strings (e.g. "2.50M" to 2500000)
parse_bdo_volume <- function(x) {
  x <- gsub(",", "", x)
  mult <- ifelse(grepl("M", x, ignore.case = TRUE), 1e6,
                 ifelse(grepl("K", x, ignore.case = TRUE), 1e3, 1))
  x_num <- as.numeric(gsub("[A-Za-z%]", "", x))
  x_num * mult
}
```

```r
# Define tickers
us_asset <- "AAPL"
ph_asset <- "BDO.PS"
crypto <- "BTC-USD"
```

Sec 4: Data Cleaning – GALEDO, Enrique Lorenzo Hermoso (data loading) and SISON, Aaron Joshua Estacio (cleaning)

Data cleaning operations: standardise dates, filter 2020-2025, count missing/duplicates, calculate daily returns.

```r
# Data Loading -- GALEDO, Enrique Lorenzo Hermoso (Sec 3: Data Acquisition)

# Load all 8 CSVs from the data directory
aapl <- read.csv(file.path(data_dir, "AAPL.csv"), stringsAsFactors = FALSE)
bdo  <- read.csv(file.path(data_dir, "BDO.csv"), stringsAsFactors = FALSE,
 ↪ fileEncoding = "UTF-8-BOM")
btc  <- read.csv(file.path(data_dir, "BTC.csv"), stringsAsFactors = FALSE)
spy  <- read.csv(file.path(data_dir, "SPY.csv"), stringsAsFactors = FALSE)
cpi      <- read.csv(file.path(data_dir, "CPI.csv"), stringsAsFactors =
 ↪ FALSE)
dgs10    <- read.csv(file.path(data_dir, "DGS10.csv"), stringsAsFactors =
 ↪ FALSE)
fedfunds <- read.csv(file.path(data_dir, "FEDFUNDS.csv"), stringsAsFactors =
 ↪ FALSE)
vix      <- read.csv(file.path(data_dir, "VIX.csv"), stringsAsFactors =
 ↪ FALSE)

# Record original observations before any processing
obs_original <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪ "VIX"),
  Original_Obs = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                   nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix))
)
```

Dataset	Original Observations
AAPL	2,512
BDO	2,441
BTC	3,652
SPY	2,514

Dataset	Original Observations
CPI	952
DGS10	16,105
FEDFUNDS	863
VIX	9,216

```
# Sec 4: Data Cleaning -- SISON, Aaron Joshua Estacio

# Standardise date formats
# AAPL, BTC, SPY: POSIXct-style strings like "2016-06-30 00:00:00-04:00"
aapl <- aapl |> mutate(Date = as.Date(ymd_hms(Date)))
btc  <- btc  |> mutate(Date = as.Date(ymd_hms(Date)))
spy  <- spy  |> mutate(Date = as.Date(ymd_hms(Date)))

# BDO: MM/DD/YYYY
bdo <- bdo |> mutate(Date = mdy(Date))

# CPI, DGS10, FEDFUNDS, VIX: YYYY-MM-DD
cpi      <- cpi      |> mutate(Date = ymd(Date))
dgs10    <- dgs10    |> mutate(Date = ymd(Date))
fedfunds <- fedfunds |> mutate(Date = ymd(Date))
vix      <- vix      |> mutate(Date = ymd(Date))

# Filter ALL datasets to 2020-01-01 to 2025-12-31
start_date <- as.Date("2020-01-01")
end_date   <- as.Date("2025-12-31")

aapl <- aapl |> filter(Date >= start_date & Date <= end_date)
bdo  <- bdo  |> filter(Date >= start_date & Date <= end_date)
btc  <- btc  |> filter(Date >= start_date & Date <= end_date)
spy  <- spy  |> filter(Date >= start_date & Date <= end_date)
cpi      <- cpi      |> filter(Date >= start_date & Date <= end_date)
dgs10    <- dgs10    |> filter(Date >= start_date & Date <= end_date)
fedfunds <- fedfunds |> filter(Date >= start_date & Date <= end_date)
vix      <- vix      |> filter(Date >= start_date & Date <= end_date)

# Observations after filtering
obs_filtered <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Filtered_Obs = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                   nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix))
)
```

```r
# Missing values count
missing_summary <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Missing_Values = c(sum(is.na(aapl)), sum(is.na(bdo)), sum(is.na(btc)),
                     sum(is.na(spy)), sum(is.na(cpi)), sum(is.na(dgs10)),
                     sum(is.na(fedfunds)), sum(is.na(vix)))
)

# Duplicate rows count
dup_summary <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX"),
  Duplicates = c(sum(duplicated(aapl)), sum(duplicated(bdo)),
   ↪  sum(duplicated(btc)),
                 sum(duplicated(spy)), sum(duplicated(cpi)),
                  ↪  sum(duplicated(dgs10)),
                 sum(duplicated(fedfunds)), sum(duplicated(vix)))
)

# Clean BDO: parse Vol. column if it exists
if ("Vol." %in% colnames(bdo)) {
  bdo <- bdo |> mutate(`Vol.` = parse_bdo_volume(`Vol.`))
}

# Calculate daily returns for OHLC assets
# Daily_Return = (Close / lag(Close)) - 1
# For BDO the Price column is the close price
aapl <- aapl |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
 ↪  - 1)
bdo  <- bdo  |> arrange(Date) |> mutate(Daily_Return = (Price / lag(Price))
 ↪  - 1)
btc  <- btc  |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
 ↪  - 1)
spy  <- spy  |> arrange(Date) |> mutate(Daily_Return = (Close / lag(Close))
 ↪  - 1)

# Note: BDO's Change. column can be used to verify Daily_Return computation
# The first observation of each asset will have NA for Daily_Return (no
 ↪  prior day)
```

Dataset	After Filtering	Missing	Duplicates
AAPL	1,508	0	0

Dataset	After Filtering	Missing	Duplicates
BDO	1,468	0	0
BTC	2,192	0	0
SPY	1,508	0	0
CPI	71	0	0
DGS10	1,500	0	0
FEDFUNDS	72	0	0
VIX	1,535	0	0

FRED data (CPI, DGS10, FEDFUNDS) has fewer rows due to monthly or business-daily reporting frequency. Missing values introduced by join operations are handled in Section 5.

Sec 5: Database Integration – SISON, Aaron Joshua Estacio

Perform joins and justify strategy.

```r
# Sec 5: Database Integration -- SISON, Aaron Joshua Estacio

# Prepare asset data for joining
# Standardise column names: BDO uses Price (close price) and Vol. (volume)
bdo_clean <- bdo |> rename(Close = Price, Volume = `Vol.`)

# Add Asset identifier to each OHLC dataset and select common columns
aapl_clean <- aapl |> mutate(Asset = "AAPL") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
bdo_clean  <- bdo_clean |> mutate(Asset = "BDO") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
btc_clean  <- btc |> mutate(Asset = "BTC") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)
spy_clean  <- spy |> mutate(Asset = "SPY") |>
  select(Date, Asset, Open, High, Low, Close, Volume, Daily_Return)

# Combine all assets into one long-format table
assets_all <- bind_rows(aapl_clean, bdo_clean, btc_clean, spy_clean)

# Pivot to wide format: one row per date, columns per asset
assets_wide <- assets_all |>
  select(Date, Asset, Close, Daily_Return) |>
  pivot_wider(names_from = Asset,
              values_from = c(Close, Daily_Return),
              names_sep = "_")

# Prepare macro datasets with descriptive column names
```

```r
cpi_clean      <- cpi      |> rename(CPI = Value)
dgs10_clean    <- dgs10    |> rename(DGS10 = Value)
fedfunds_clean <- fedfunds |> rename(FEDFUNDS = Value)
vix_clean      <- vix      |> rename(VIX = Value)



# Demonstration of all four join types

# 1) left_join: keep all asset dates, attach macro data where available
left_example <- assets_wide |>
  left_join(cpi_clean, by = "Date") |>
  left_join(dgs10_clean, by = "Date") |>
  left_join(fedfunds_clean, by = "Date") |>
  left_join(vix_clean, by = "Date")

# 2) inner_join: only dates present in ALL datasets
inner_example <- assets_wide |>
  inner_join(cpi_clean, by = "Date") |>
  inner_join(dgs10_clean, by = "Date") |>
  inner_join(fedfunds_clean, by = "Date") |>
  inner_join(vix_clean, by = "Date")

# 3) full_join: all dates from any dataset
full_example <- assets_wide |>
  full_join(cpi_clean, by = "Date") |>
  full_join(dgs10_clean, by = "Date") |>
  full_join(fedfunds_clean, by = "Date") |>
  full_join(vix_clean, by = "Date")

# 4) anti_join: asset dates with NO macro data
anti_example <- assets_wide |>
  anti_join(cpi_clean, by = "Date")

# Build the final analytical dataset
final_dataset <- left_example



# Observation reduction table

reduction_table <- data.frame(
  Dataset = c("AAPL", "BDO", "BTC", "SPY", "CPI", "DGS10", "FEDFUNDS",
   ↪  "VIX",
              "Assets (wide)", "Final Dataset"),
```

```
  Original_Obs = c(obs_original$Original_Obs, NA, NA),
  After_Cleaning = c(nrow(aapl), nrow(bdo), nrow(btc), nrow(spy),
                     nrow(cpi), nrow(dgs10), nrow(fedfunds), nrow(vix),
                     nrow(assets_wide), NA),
  After_Joins = c(rep(NA, 8), NA, nrow(final_dataset))
)
```

Sec 6: SQL-Based Financial Investigation – SISON, Aaron Joshua Estacio (Q1-Q3, Q8-Q10)

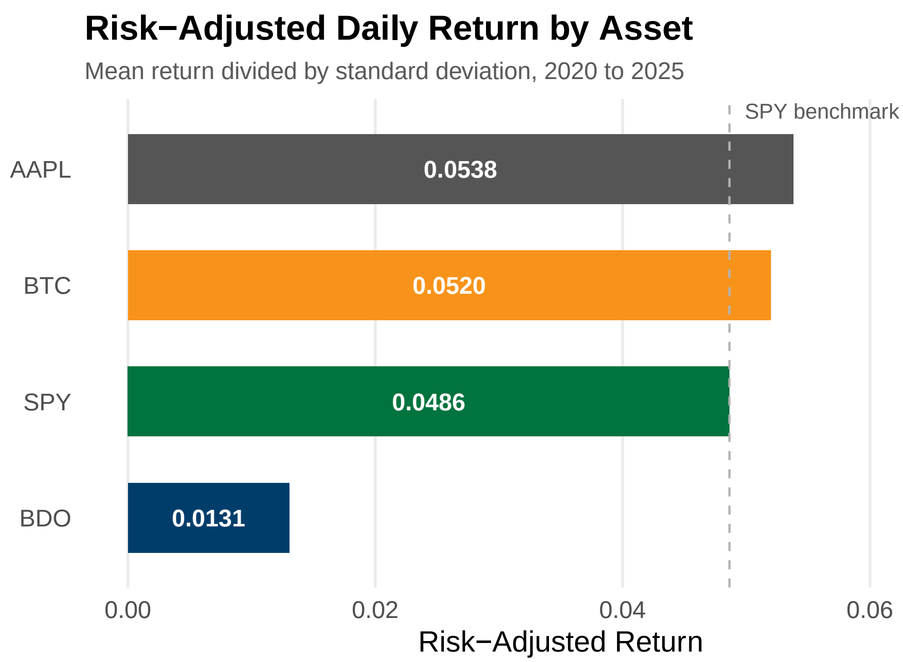
10 meaningful financial questions using select, filter, mutate, arrange, group_by, summarise.

```
# SQLite database setup for SQL-based investigations

con <- dbConnect(RSQLite::SQLite(), ":memory:")

# Build long-format analytical dataset for SQL queries
# Each row is one asset-date combination with macro variables attached
sql_long <- bind_rows(
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_AAPL, Daily_Return = Daily_Return_AAPL) |>
    mutate(Asset = "AAPL"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_BDO, Daily_Return = Daily_Return_BDO) |>
    mutate(Asset = "BDO"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_BTC, Daily_Return = Daily_Return_BTC) |>
    mutate(Asset = "BTC"),
  final_dataset |>
    select(Date, CPI, DGS10, FEDFUNDS, VIX,
           Close = Close_SPY, Daily_Return = Daily_Return_SPY) |>
    mutate(Asset = "SPY")
)

# Pre-calculate CPI YoY for Q10 (inflation analysis)
# CPI is monthly; YoY change = (current / 12-month-ago) - 1
cpi_yoy <- cpi_clean |>
  arrange(Date) |>
  mutate(CPI_YoY = (CPI / lag(CPI, 12)) - 1) |>
  select(Date, CPI_YoY)
```

```
# Join CPI_YoY into the long dataset and forward-fill to cover all trading
↪ days
sql_long <- sql_long |>
  left_join(cpi_yoy, by = "Date") |>
  arrange(Date) |>
  tidyr::fill(CPI_YoY, .direction = "down")


dbWriteTable(con, "integrated_data", sql_long, overwrite = TRUE)
fin_data <- tbl(con, "integrated_data")
```

```
# Q1: Risk-Adjusted Daily Return by Asset -- SISON, Aaron Joshua Estacio

q1 <- fin_data |>
  group_by(Asset) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE)
  ) |>
  mutate(Risk_Adjusted_Return = Mean_Daily_Return / SD_Daily_Return) |>
  arrange(desc(Risk_Adjusted_Return)) |>
  collect()

spy_ratio <- q1 |> filter(Asset == "SPY") |> pull(Risk_Adjusted_Return)

ggplot(q1, aes(x = Risk_Adjusted_Return, y = reorder(Asset,
↪ Risk_Adjusted_Return),
                fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_vline(xintercept = spy_ratio, linetype = "dashed",
             color = "#B3B3B3", linewidth = 0.4) +
  geom_text(aes(x = Risk_Adjusted_Return / 2,
                label = sprintf("%.4f", Risk_Adjusted_Return)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  annotate("text", x = spy_ratio, y = 4.5, label = "SPY benchmark",
           size = 2.8, color = "#595959", hjust = -0.1) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Risk-Adjusted Daily Return by Asset",
       subtitle = "Mean return divided by standard deviation, 2020 to 2025",
       x = "Risk-Adjusted Return", y = NULL) +
  xlim(0, 0.07) +
  theme_minimal(base_size = 11) +
```

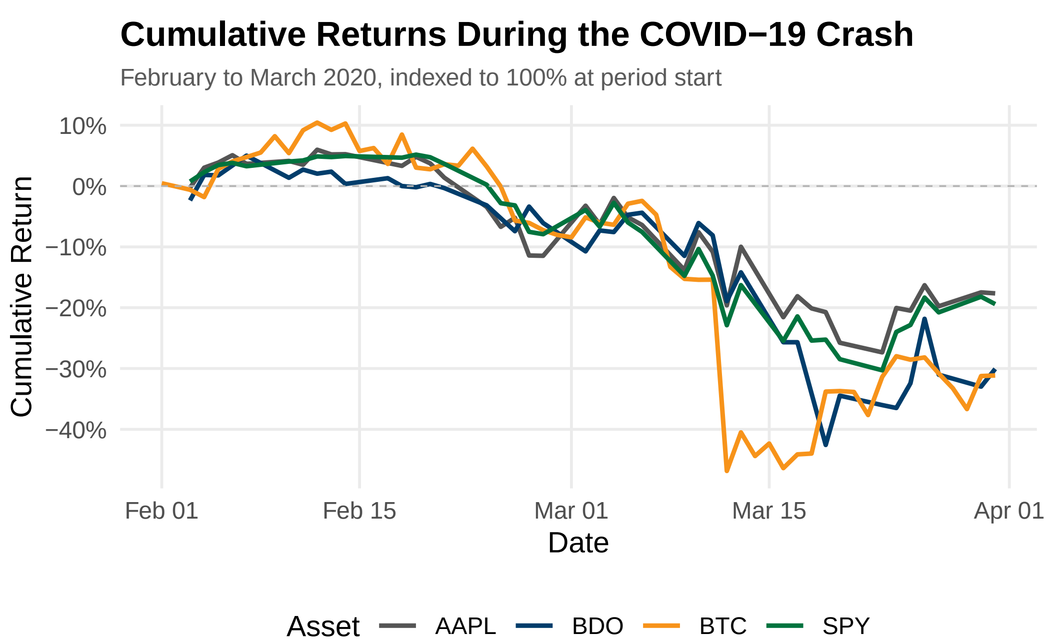
```
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/fig_q1.pdf", width = 7, height = 3.5, device = "pdf")
```

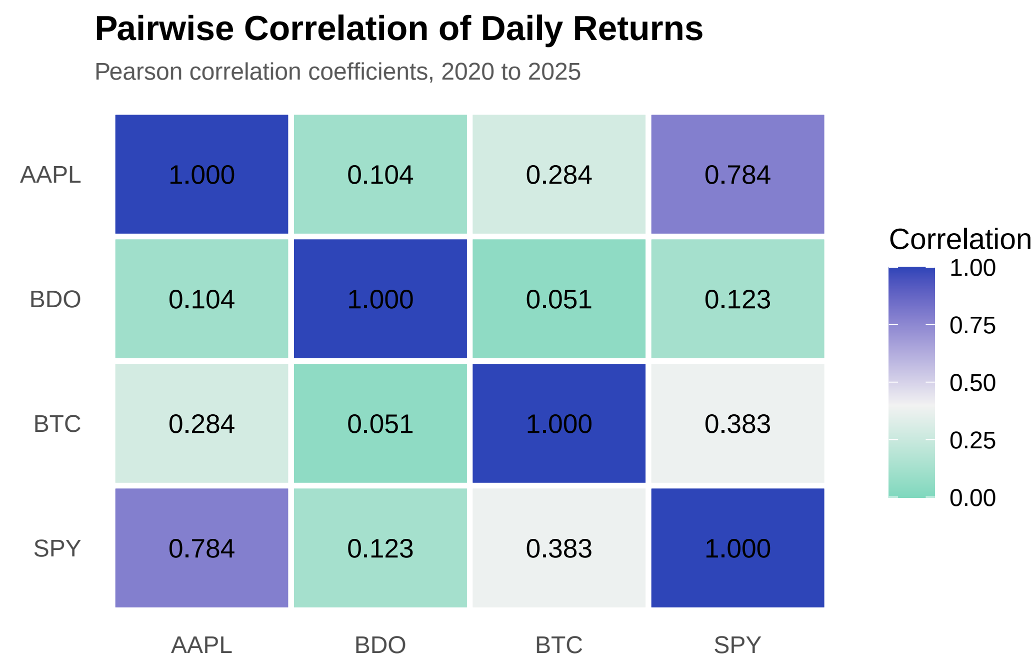
```
# Q2: COVID-19 Crash Cumulative Returns -- SISON, Aaron Joshua Estacio

q2_raw <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  mutate(Date = as.Date(Date))

q2 <- q2_raw |>
  filter(Date >= as.Date("2020-02-01") & Date <= as.Date("2020-03-31")) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1)

q2_summary <- q2 |>
  summarise(
    Start_Date = min(Date),
```

```
    End_Date = max(Date),
    Total_Cumulative_Return = last(Cumulative_Return),
    Observations = n()
  )

ggplot(q2, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
   ↪  linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Cumulative Returns During the COVID-19 Crash",
       subtitle = "February to March 2020, indexed to 100% at period start",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```




```
ggsave("../reports/fig_q2.pdf", width = 7, height = 4, device = "pdf")
```

```r
# Q3: Pairwise Correlation of Daily Returns -- SISON, Aaron Joshua Estacio

returns_wide <- fin_data |>
  select(Date, Asset, Daily_Return) |>
  collect() |>
  pivot_wider(names_from = Asset, values_from = Daily_Return)

cor_matrix <- returns_wide |>
  select(AAPL, BDO, BTC, SPY) |>
  cor(use = "pairwise.complete.obs")

cor_long <- as.data.frame(cor_matrix) |>
  tibble::rownames_to_column("Asset1") |>
  tidyr::pivot_longer(-Asset1, names_to = "Asset2", values_to =
↪  "Correlation") |>
  mutate(Asset1 = factor(Asset1, levels = rev(c("AAPL", "BDO", "BTC",
   ↪  "SPY"))),
         Asset2 = factor(Asset2, levels = c("AAPL", "BDO", "BTC", "SPY")))

ggplot(cor_long, aes(x = Asset2, y = Asset1, fill = Correlation)) +
  geom_tile(color = "white", linewidth = 1) +
  geom_text(aes(label = sprintf("%.3f", Correlation)), size = 3.5) +
  scale_fill_gradient2(low = "#1DC9A4", mid = "#F2F2F2", high = "#2E45B8",
                       midpoint = 0.4, limits = c(0, 1)) +
  labs(title = "Pairwise Correlation of Daily Returns",
       subtitle = "Pearson correlation coefficients, 2020 to 2025",
       x = NULL, y = NULL, fill = "Correlation") +
  theme_minimal(base_size = 11) +
  theme(panel.grid = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "right")
```

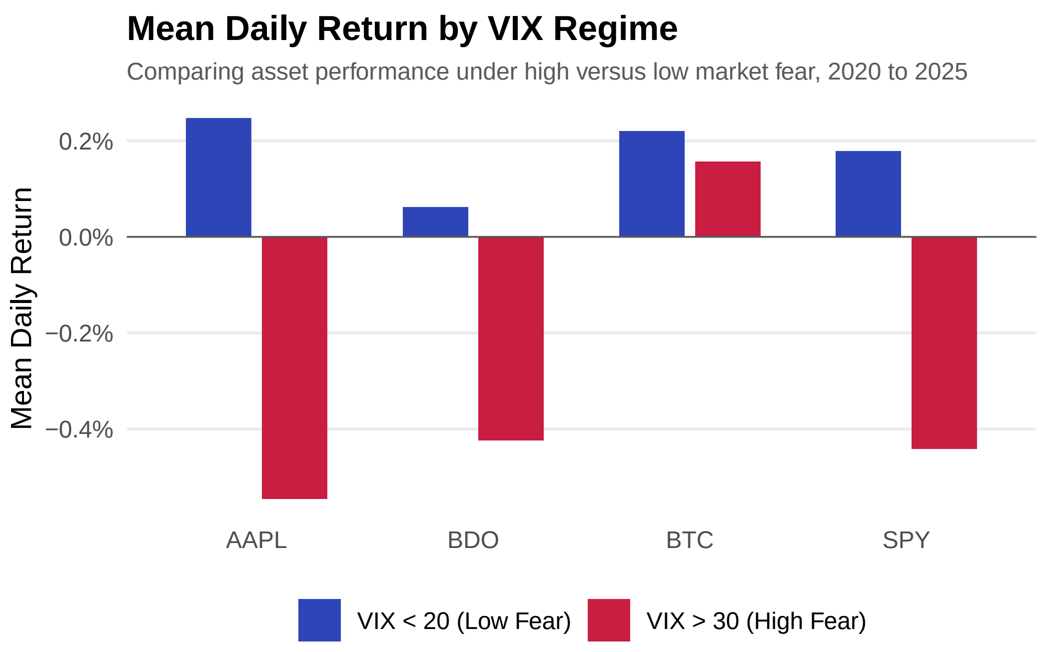



```r
ggsave("../reports/fig_q3.pdf", width = 6, height = 4.5, device = "pdf")
```

```r
# Q4-Q7: Assigned to CRUZ, Ricardo Miguel Inigo
# Q4: Asset Ranking by Total Return
# Q5: Best and Worst Trading Days
# Q6: Moving Average Crossover Analysis
# Q7: Volume and Volatility Relationship
# Code to be provided by responsible group member
```

```r
# Q8: VIX Regime Analysis -- SISON, Aaron Joshua Estacio

q8_raw <- fin_data |>
  filter(!is.na(VIX)) |>
  select(Asset, Date, Daily_Return, VIX) |>
  collect() |>
  mutate(Date = as.Date(Date))

q8 <- q8_raw |>
  mutate(
    VIX_Regime = case_when(
      VIX > 30 ~ "High Volatility (VIX > 30)",
      VIX < 20 ~ "Low Volatility (VIX < 20)",
      TRUE ~ "Moderate (20 <= VIX <= 30)"
    )
  ) |>
```

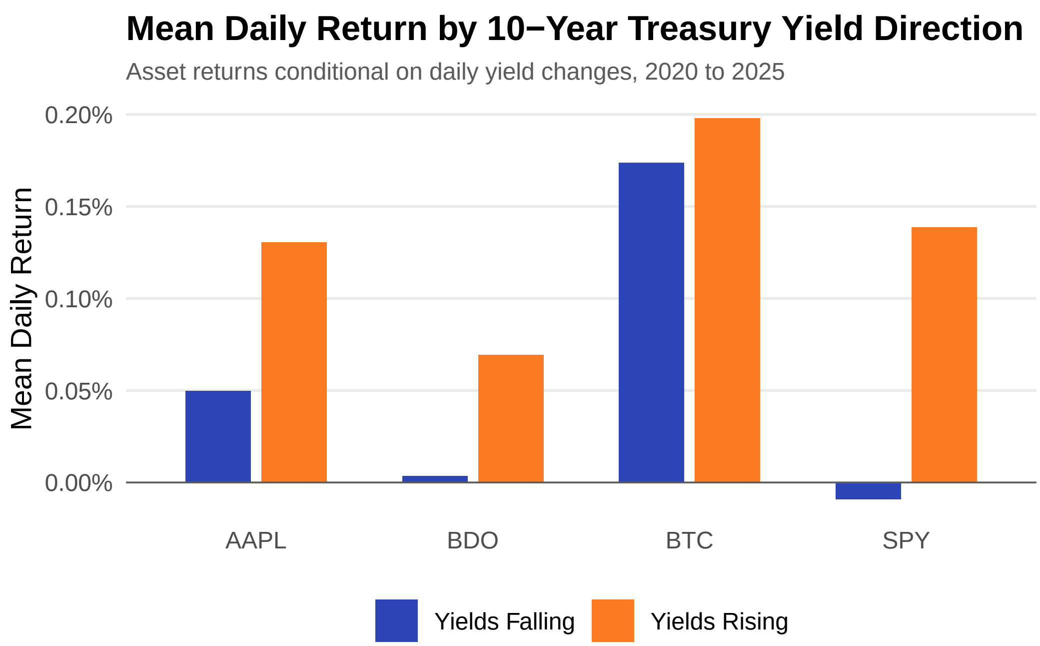
```r
  group_by(Asset, VIX_Regime) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, VIX_Regime)

q8_plot <- q8 |>
  filter(VIX_Regime %in% c("High Volatility (VIX > 30)", "Low Volatility
  ↪  (VIX < 20)")) |>
  mutate(VIX_Regime = recode(VIX_Regime,
    "High Volatility (VIX > 30)" = "VIX > 30 (High Fear)",
    "Low Volatility (VIX < 20)" = "VIX < 20 (Low Fear)"))

ggplot(q8_plot, aes(x = Asset, y = Mean_Daily_Return, fill = VIX_Regime)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("VIX > 30 (High Fear)" = "#C91D42",
                               "VIX < 20 (Low Fear)" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by VIX Regime",
       subtitle = "Comparing asset performance under high versus low market
       ↪  fear, 2020 to 2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```




```
ggsave("../reports/fig_q8.pdf", width = 7, height = 4, device = "pdf")
```

```
# Q9: Interest Rate Sensitivity -- SISON, Aaron Joshua Estacio

q9_raw <- fin_data |>
  filter(!is.na(DGS10)) |>
  select(Asset, Date, Daily_Return, DGS10) |>
  collect() |>
  mutate(Date = as.Date(Date))

q9 <- q9_raw |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(DGS10_Change = DGS10 - lag(DGS10)) |>
  ungroup() |>
  filter(!is.na(DGS10_Change)) |>
  mutate(
    Yield_Direction = case_when(
      DGS10_Change > 0 ~ "Yields Rising",
      DGS10_Change < 0 ~ "Yields Falling",
      TRUE ~ "No Change"
    )
  ) |>
  group_by(Asset, Yield_Direction) |>
  summarise(
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
```

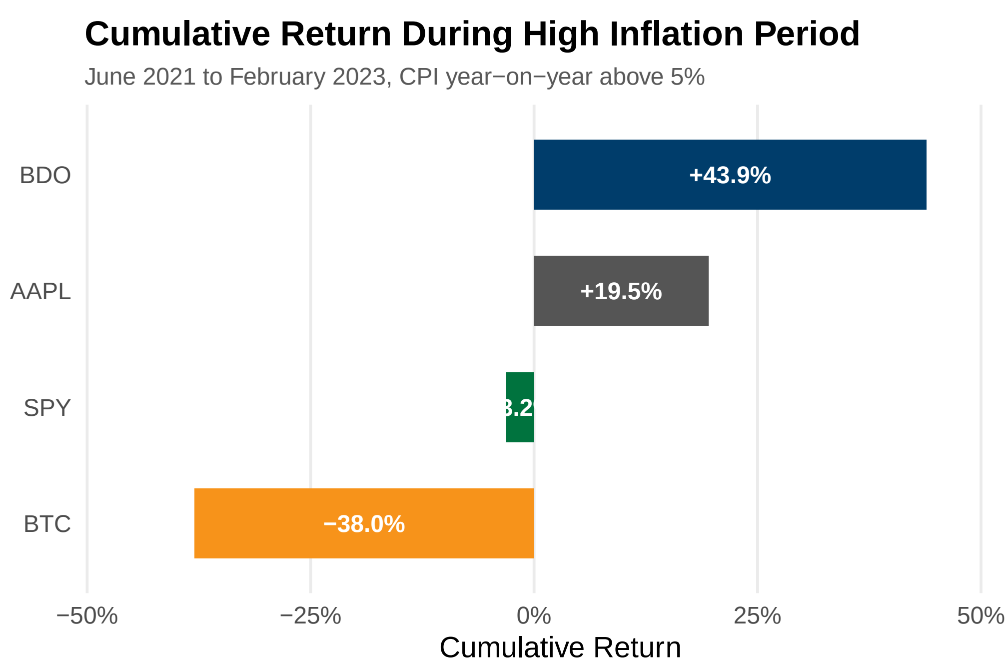
```
    SD_Daily_Return = sd(Daily_Return, na.rm = TRUE),
    Observations = n(),
    .groups = "drop"
  ) |>
  arrange(Asset, Yield_Direction)

q9_plot <- q9 |>
  filter(Yield_Direction %in% c("Yields Rising", "Yields Falling"))

ggplot(q9_plot, aes(x = Asset, y = Mean_Daily_Return, fill =
 ↪  Yield_Direction)) +
  geom_col(position = position_dodge(width = 0.7), width = 0.6) +
  geom_hline(yintercept = 0, linewidth = 0.3, color = "#595959") +
  scale_fill_manual(values = c("Yields Rising" = "#F97A1F",
                               "Yields Falling" = "#2E45B8")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Mean Daily Return by 10-Year Treasury Yield Direction",
       subtitle = "Asset returns conditional on daily yield changes, 2020 to
        ↪  2025",
       x = NULL, y = "Mean Daily Return", fill = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```




```r
ggsave("../reports/fig_q9.pdf", width = 7, height = 4, device = "pdf")
```

```r
# Q10: Inflation Hedge Performance -- SISON, Aaron Joshua Estacio

q10_raw <- fin_data |>
  select(Date, Asset, Daily_Return, CPI_YoY) |>
  collect() |>
  mutate(Date = as.Date(Date))

q10 <- q10_raw |>
  filter(!is.na(CPI_YoY) & CPI_YoY > 0.05) |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  summarise(
    Start_Date = min(Date),
    End_Date = max(Date),
    Start_CPI_YoY = first(CPI_YoY),
    End_CPI_YoY = last(CPI_YoY),
    Total_Cumulative_Return = last(Cumulative_Return),
    Mean_Daily_Return = mean(Daily_Return, na.rm = TRUE),
    Observations = n()
  ) |>
  arrange(desc(Total_Cumulative_Return))
```

```r
q10_plot <- q10 |>
  mutate(Asset = factor(Asset, levels = q10$Asset[order(q10$Total_Cumulative_Return)]))

ggplot(q10_plot, aes(x = Total_Cumulative_Return, y = Asset, fill = Asset)) +
  geom_col(width = 0.6, show.legend = FALSE) +
  geom_text(aes(x = Total_Cumulative_Return / 2,
                label = sprintf("%+.1f%%", Total_Cumulative_Return * 100)),
            hjust = 0.5, size = 3.2, color = "white", fontface = "bold") +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format(),
                     expand = expansion(mult = c(0.15, 0.15))) +
  labs(title = "Cumulative Return During High Inflation Period",
       subtitle = "June 2021 to February 2023, CPI year-on-year above 5%",
       x = "Cumulative Return", y = NULL) +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.y = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```



```r
ggsave("../reports/fig_q10.pdf", width = 7, height = 3.5, device = "pdf")
```

Sec 7: Descriptive Statistics – GALEDO, Enrique Lorenzo Hermoso

Mean, median, variance, std dev, min, max, range, skewness, kurtosis.

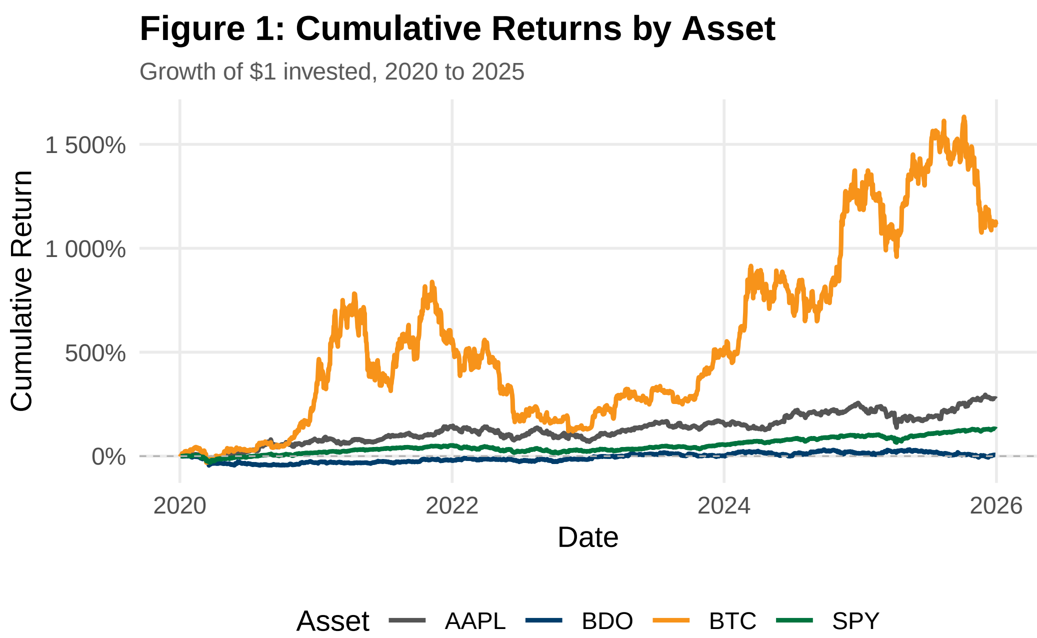
```r
# Sec 7: Descriptive Statistics -- compute mean, median, variance, skewness,
↪ kurtosis -- GALEDO, Enrique Lorenzo Hermoso
```

Sec 8: Data Visualization – SEBALLOS, Josiah Dweyn Panganiban

At least 8 professional-quality figures.

```r
# Sec 8: Data Visualization -- create 8 professional figures with ggplot2 --
↪ SEBALLOS, Josiah Dweyn Panganiban
```

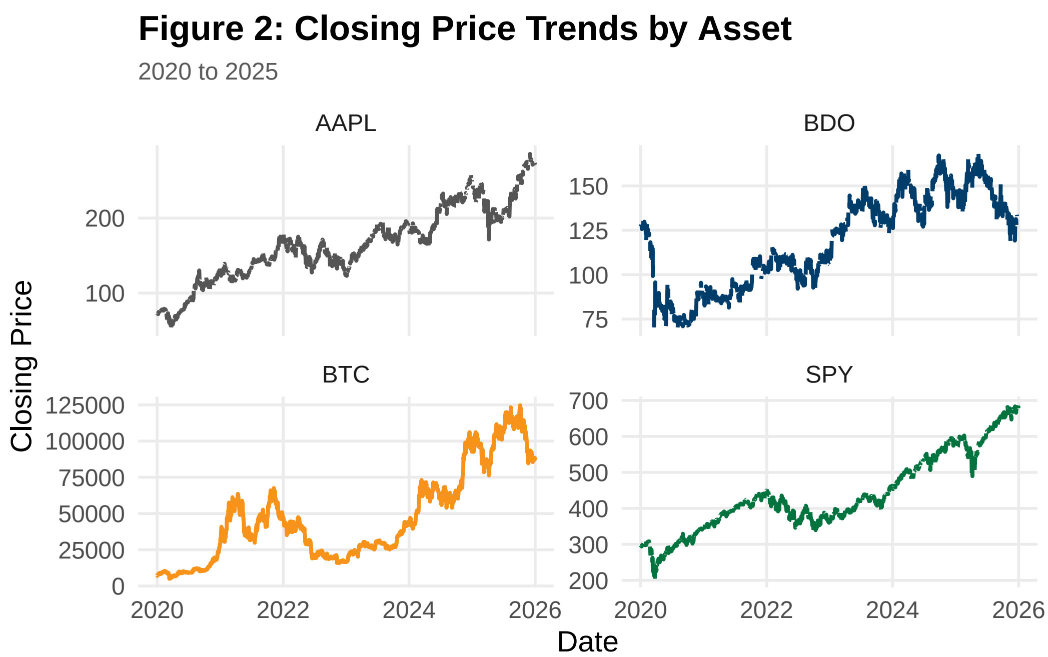
```r
# Figure 1: Cumulative Returns by Asset
fig1_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Cumulative_Return = cumprod(1 + Daily_Return) - 1) |>
  ungroup()

ggplot(fig1_data, aes(x = Date, y = Cumulative_Return, color = Asset)) +
  geom_line(linewidth = 0.8) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "#B3B3B3",
  ↪ linewidth = 0.3) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Figure 1: Cumulative Returns by Asset",
       subtitle = "Growth of $1 invested, 2020 to 2025",
       x = "Date", y = "Cumulative Return", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```




```r
ggsave("../reports/fig_viz1.pdf", width = 7, height = 4, device = "pdf")
```

Figure 1: Cumulative Returns by Asset overlays each asset's daily return distribution against a fitted normal curve. AAPL is the only asset with positive skewness (0.28), meaning its rare extreme days lean toward the upside. BTC and BDO both show negative skewness (-0.48 and -0.40 respectively), indicating tail risk weighted toward sudden losses. SPY records the highest kurtosis of all four assets (12.97) despite a comparatively mild skew (-0.26), meaning its returns are usually tightly clustered near zero but still capable of rare, extreme shock days. All four kurtosis values exceed 3, the benchmark for a normal distribution, confirming that none of these assets follow a standard bell curve.

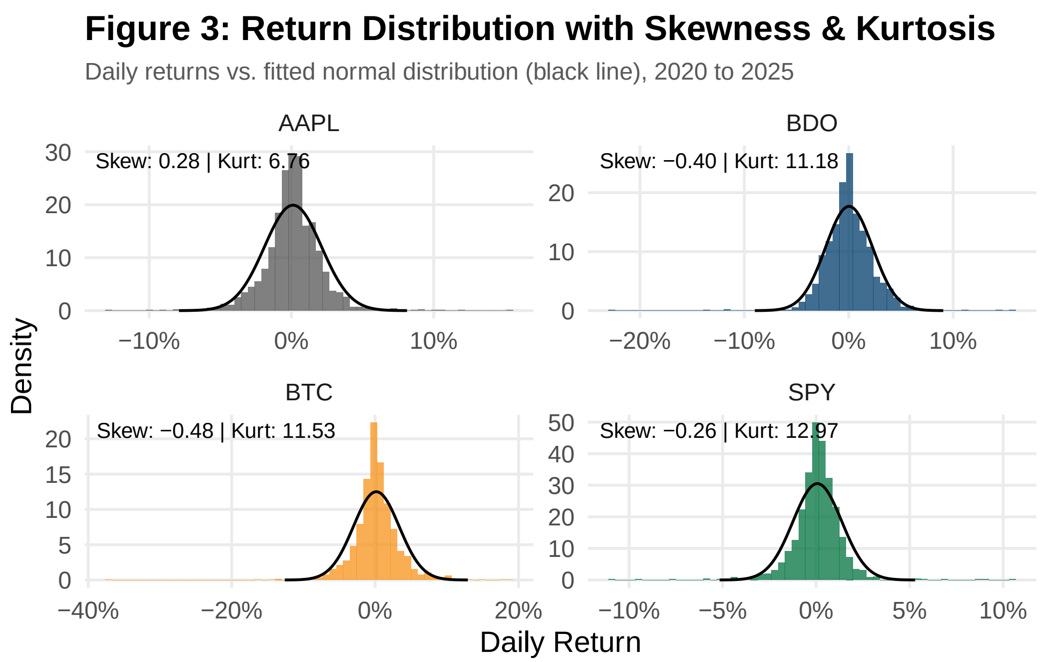
```r
# Figure 2: Closing Price Trends by Asset
ggplot(sql_long, aes(x = Date, y = Close, color = Asset)) +
  geom_line(linewidth = 0.7, show.legend = FALSE, na.rm = TRUE) +
  facet_wrap(~ Asset, scales = "free_y", ncol = 2) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  labs(title = "Figure 2: Closing Price Trends by Asset",
       subtitle = "2020 to 2025",
       x = "Date", y = "Closing Price") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/fig_viz2.pdf", width = 7, height = 5, device = "pdf")
```

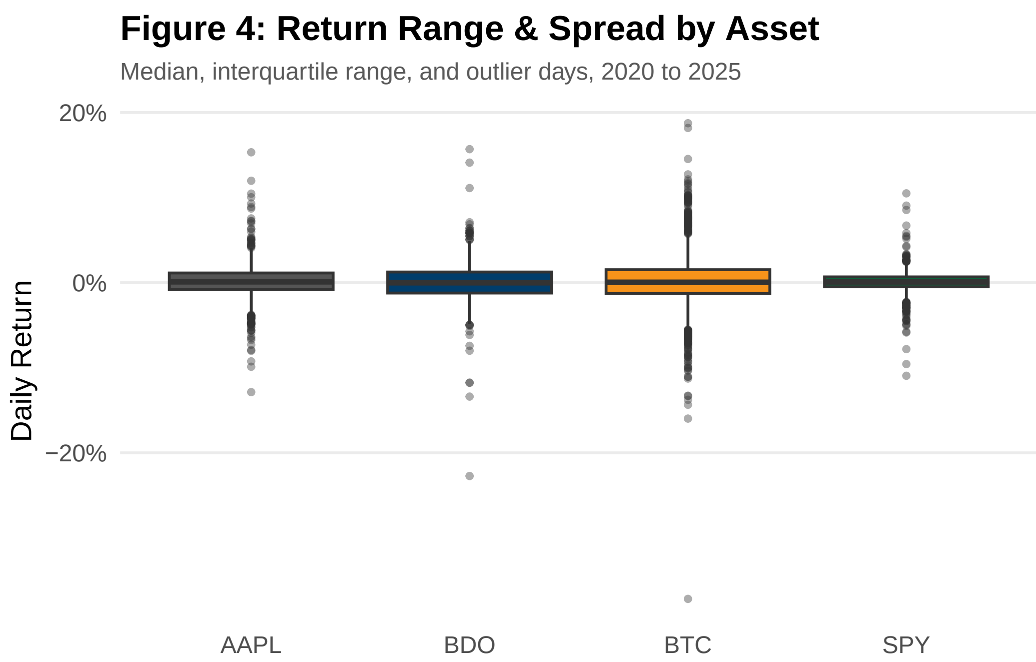
```r
# Figure 3: Return Distribution with Skewness & Kurtosis
fig3_data <- sql_long |> filter(!is.na(Daily_Return))

# Manual skewness/kurtosis (no extra package needed)
calc_skew <- function(x) {
  n <- length(x); m <- mean(x)
  (sum((x - m)^3) / n) / (sd(x)^3)
}
calc_kurt <- function(x) {
  n <- length(x); m <- mean(x)
  (sum((x - m)^4) / n) / (sd(x)^2)^2 - 3
}

fig3_stats <- fig3_data |>
  group_by(Asset) |>
  summarise(
    Mean = mean(Daily_Return), SD = sd(Daily_Return),
    Skewness = calc_skew(Daily_Return), Kurtosis = calc_kurt(Daily_Return),
    .groups = "drop"
  ) |>
  mutate(label = sprintf("Skew: %.2f | Kurt: %.2f", Skewness, Kurtosis))

# Build normal-curve overlay per asset using each asset's own mean/SD
normal_curves <- fig3_stats |>
```

```r
  rowwise() |>
  reframe(
    Asset = Asset,
    x = seq(Mean - 4 * SD, Mean + 4 * SD, length.out = 200),
    y = dnorm(x, mean = Mean, sd = SD)
  )

ggplot(fig3_data, aes(x = Daily_Return)) +
  geom_histogram(aes(y = after_stat(density), fill = Asset), bins = 60,
                 alpha = 0.75, show.legend = FALSE) +
  geom_line(data = normal_curves, aes(x = x, y = y), color = "black",
   ↪  linewidth = 0.5) +
  geom_text(data = fig3_stats, aes(x = -Inf, y = Inf, label = label),
            hjust = -0.05, vjust = 1.5, size = 2.8) +
  facet_wrap(~ Asset, scales = "free", ncol = 2) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_x_continuous(labels = scales::percent_format()) +
  labs(title = "Figure 3: Return Distribution with Skewness & Kurtosis",
       subtitle = "Daily returns vs. fitted normal distribution (black
        ↪  line), 2020 to 2025",
       x = "Daily Return", y = "Density") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/fig_viz3.pdf", width = 7, height = 5, device = "pdf")
```

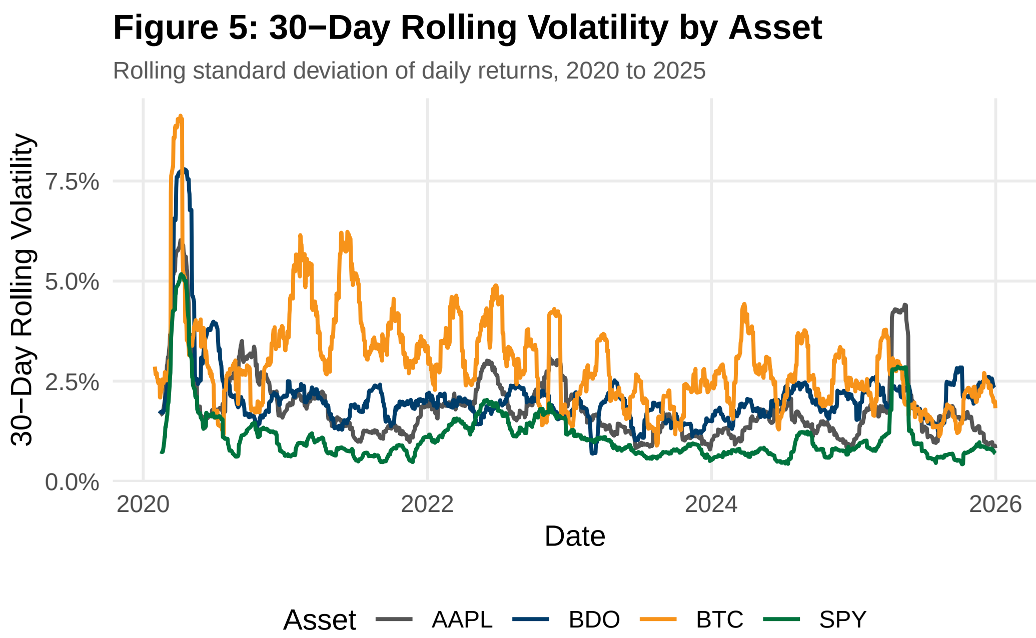
```
# Figure 4: Return Range & Spread by Asset
ggplot(fig3_data, aes(x = Asset, y = Daily_Return, fill = Asset)) +
  geom_boxplot(show.legend = FALSE, outlier.size = 0.8, outlier.alpha = 0.4)
   ↪  +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                               BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Figure 4: Return Range & Spread by Asset",
       subtitle = "Median, interquartile range, and outlier days, 2020 to
        ↪  2025",
       x = NULL, y = "Daily Return") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```
ggsave("../reports/fig_viz4.pdf", width = 7, height = 4, device = "pdf")
```

```
# Figure 5: 30-Day Rolling Volatility by Asset
rolling_sd <- function(x, n = 30) {
  sapply(seq_along(x), function(i) {
    if (i < n) return(NA)
    sd(x[(i - n + 1):i], na.rm = TRUE)
  })
}

fig5_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Rolling_Vol = rolling_sd(Daily_Return, 30)) |>
  ungroup() |>
  filter(!is.na(Rolling_Vol))

ggplot(fig5_data, aes(x = Date, y = Rolling_Vol, color = Asset)) +
  geom_line(linewidth = 0.7) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Figure 5: 30-Day Rolling Volatility by Asset",
       subtitle = "Rolling standard deviation of daily returns, 2020 to
        ↪  2025",
```

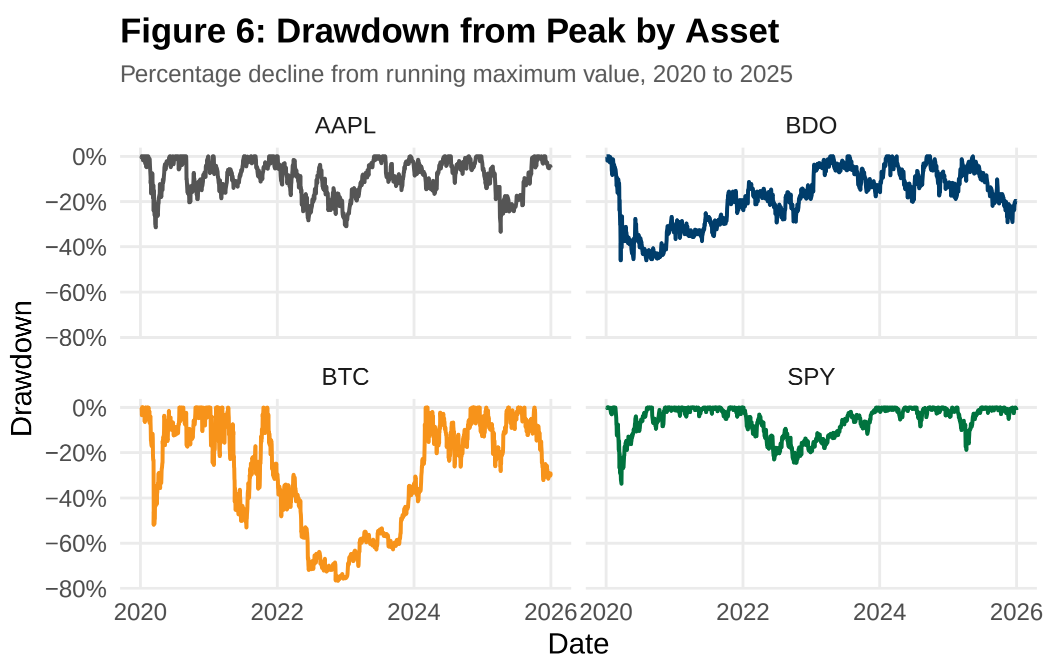
```
      x = "Date", y = "30-Day Rolling Volatility", color = "Asset") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"),
        legend.position = "bottom")
```




```
ggsave("../reports/fig_viz5.pdf", width = 7, height = 4, device = "pdf")
```

```
# Figure 6: Drawdown from Peak by Asset
fig6_data <- fig1_data |>
  group_by(Asset) |>
  arrange(Date) |>
  mutate(Running_Max = cummax(1 + Cumulative_Return),
         Drawdown = (1 + Cumulative_Return) / Running_Max - 1) |>
  ungroup()

ggplot(fig6_data, aes(x = Date, y = Drawdown, color = Asset)) +
  geom_line(linewidth = 0.7) +
  facet_wrap(~ Asset, ncol = 2) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Figure 6: Drawdown from Peak by Asset",
```

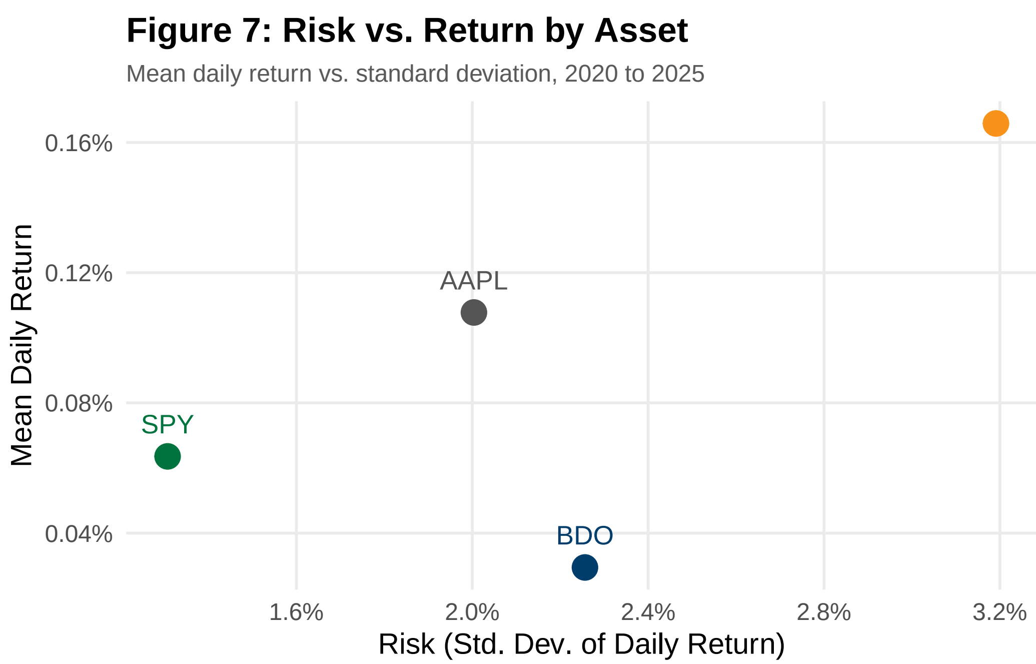
```r
       subtitle = "Percentage decline from running maximum value, 2020 to
        ↪  2025",
       x = "Date", y = "Drawdown") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        legend.position = "none",
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/fig_viz6.pdf", width = 7, height = 5, device = "pdf")
```

```r
# Figure 7: Risk vs. Return by Asset
fig7_data <- sql_long |>
  filter(!is.na(Daily_Return)) |>
  group_by(Asset) |>
  summarise(Mean_Return = mean(Daily_Return), SD_Return = sd(Daily_Return),
   ↪  .groups = "drop")

ggplot(fig7_data, aes(x = SD_Return, y = Mean_Return, color = Asset)) +
  geom_point(size = 4) +
  geom_text(aes(label = Asset), vjust = -1.2, size = 3.5, show.legend =
   ↪  FALSE) +
  scale_color_manual(values = c(AAPL = "#555555", BDO = "#003D6B",
                                BTC = "#F7931A", SPY = "#00733E")) +
```

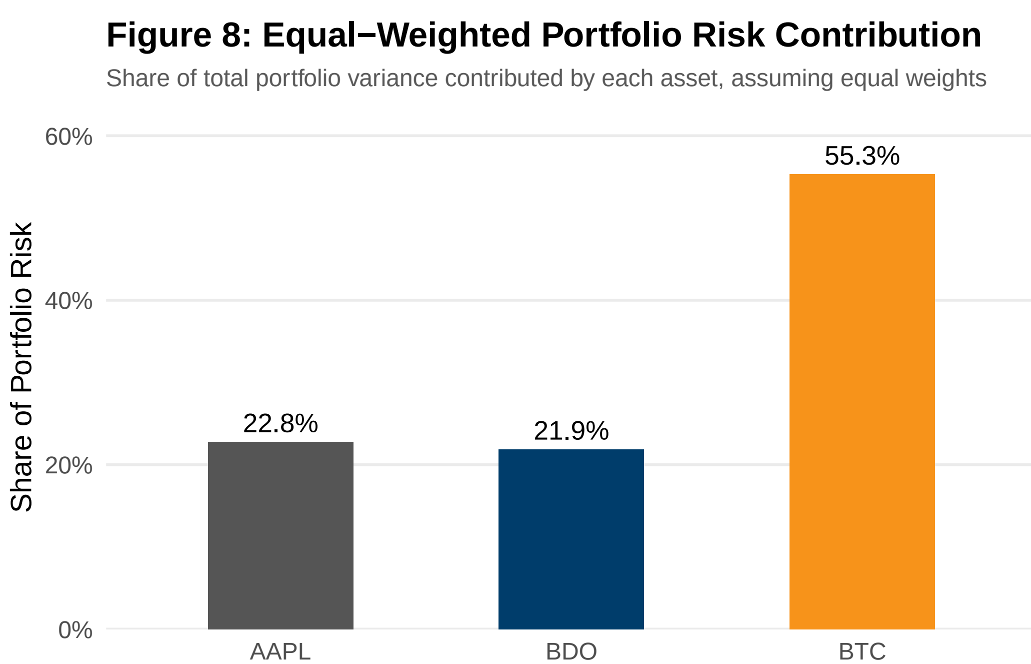
```r
  scale_x_continuous(labels = scales::percent_format()) +
  scale_y_continuous(labels = scales::percent_format()) +
  labs(title = "Figure 7: Risk vs. Return by Asset",
       subtitle = "Mean daily return vs. standard deviation, 2020 to 2025",
       x = "Risk (Std. Dev. of Daily Return)", y = "Mean Daily Return") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        legend.position = "none",
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```




```r
ggsave("../reports/fig_viz7.pdf", width = 7, height = 5, device = "pdf")
```

```r
# Figure 8: Equal-Weighted Portfolio Risk Contribution
returns_matrix <- sql_long |>
  filter(Asset %in% c("AAPL", "BDO", "BTC"), !is.na(Daily_Return)) |>
  select(Date, Asset, Daily_Return) |>
  pivot_wider(names_from = Asset, values_from = Daily_Return) |>
  select(AAPL, BDO, BTC) |>
  na.omit()

cov_matrix <- cov(returns_matrix)
weights <- c(AAPL = 1/3, BDO = 1/3, BTC = 1/3)
port_var <- as.numeric(t(weights) %*% cov_matrix %*% weights)
```

```
marginal_contrib <- cov_matrix %*% weights
risk_contrib <- as.numeric(weights * marginal_contrib / port_var)

fig8_data <- data.frame(Asset = names(weights), Risk_Contribution =
 ↪  risk_contrib)

ggplot(fig8_data, aes(x = Asset, y = Risk_Contribution, fill = Asset)) +
  geom_col(width = 0.5, show.legend = FALSE) +
  geom_text(aes(label = scales::percent(Risk_Contribution, accuracy = 0.1)),
            vjust = -0.5, size = 3.5) +
  scale_fill_manual(values = c(AAPL = "#555555", BDO = "#003D6B", BTC =
   ↪  "#F7931A")) +
  scale_y_continuous(labels = scales::percent_format(), expand =
   ↪  expansion(mult = c(0, 0.15))) +
  labs(title = "Figure 8: Equal-Weighted Portfolio Risk Contribution",
       subtitle = "Share of total portfolio variance contributed by each
        ↪  asset, assuming equal weights",
       x = NULL, y = "Share of Portfolio Risk") +
  theme_minimal(base_size = 11) +
  theme(panel.grid.major.x = element_blank(),
        panel.grid.minor = element_blank(),
        axis.ticks = element_blank(),
        plot.title = element_text(face = "bold", size = 13),
        plot.subtitle = element_text(size = 9, color = "#595959"))
```



```r
ggsave("../reports/fig_viz8.pdf", width = 7, height = 4, device = "pdf")
```

Sec 9: Exploratory Analysis – SEECHUNG, Camille Castro

Compare returns, volatility, inflationary periods, high-VIX periods, interest-rate environments.

```r
# Sec 9: Exploratory Financial Analysis -- compare returns, volatility,
↪  inflation, VIX, interest rates -- SEECHUNG, Camille Castro
```

Sec 10: Investment Recommendation – SEECHUNG, Camille Castro

Allocate USD 1,000,000.

```r
# Sec 10: Investment Recommendation -- USD 1,000,000 allocation among AAPL,
↪  BDO, BTC, cash -- SEECHUNG, Camille Castro
```

Sec 11: Executive Dashboard – CRUZ, Ricardo Miguel Iñigo

```r
# Sec 11: Executive Dashboard -- key stats, charts, SQL findings,
↪  recommendation summary -- CRUZ, Ricardo Miguel Iñigo
```